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If You Tweet, They Will Follow: CEO Tweets, Social Capital, and Investor Say-on-Pay Judgments

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ABSTRACT: Chief executive officers (CEOs) are increasingly using social media to disclose information and communicate with investors. Although findings from archival research show some benefits to social media use, little is known about how the social nature of this disclosure channel affects individual investors. Accordingly, we develop a mediation model based on social capital theory that predicts social media disclosure channels lead investors to perceive enhanced feelings of connectedness (i.e., social capital) with the CEO, resulting in more positive judgments of the CEO. Specifically, our model predicts CEO disclosures via Twitter, versus web-based disclosures, lead to enhanced perceptions of social capital, which in turn positively impact investor recommendations for CEO compensation. Using an experiment with individual investors, our results provide robust support for our theoretical mediation model. Our findings enhance our understanding of how CEOs' social media use influences individual investors and have implications for standard setters, investors, and firms.

Keywords: Twitter; parasocial interaction; social capital theory; say-on-pay.

I. INTRODUCTION

Firms and chief executive officers (CEOs) are increasingly using social media to disclose information and communicate with investors. Social media, such as Twitter, enable timelier disclosure to a broader audience than alternative disclosure channels (e.g., paper- and web-based). Regulators have endorsed corporate use of social media contending that such use will improve the overall information environment, ultimately improving investor decision making (SEC 2013). Consistent with these claims, research provides evidence that a firm's use of social media is associated with reductions in information asymmetry (Saxton and Anker 2013; Blankespoor, Miller, and White 2014) and improvements in firm performance (Du and Jiang 2015). To further improve communication and firm visibility, CEOs are also using individual (versus corporate) social media accounts to engage directly with investors (Alexander and Gentry 2014). CEOs report using social media to improve their reputation, enhance relationships with stakeholders, and improve business results (Weber Shandwick 2014). However, research is lacking about how such use affects individual investors. In this study, we explore the social nature of this disclosure channel and investigate the psychological processes underlying individual investors' acquisition of a CEO's disclosures via social media.

It is important to understand how social media affects individual investors for at least two primary reasons. First, the rise in shareholder activism has resulted in managers being held more accountable to their firms' investors, particularly in regard to issues involving executive compensation (Gillan and Starks 2000, 2007; Goranova and Ryan 2014). Individual investors own

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approximately 35 percent of equity investments in U.S. public companies (Broadridge and PwC 2018) and are one of the most active investor groups in terms of the number of shareholder proposals submitted annually (Gillan and Starks 2007). However, little is known about how individual investors evaluate compensation issues due to a lack of heterogeneity in research methods (i.e., archival-based research dominates the field of study) (Obermann and Velte 2018). Given individual investors' impact on firms and management, it is important to better understand their decision-making processes. Second, the investment-related judgment and decision-making processes of individual investors differ from other investor groups. For example, individual investors are more prone to using heuristic processing when making investment decisions (Maines and McDaniel 2000), such as being influenced by varying disclosure channels. As such, the SEC (2000) has expressed concern with the influence of various disclosure channels on individual investors. These concerns are important as the SEC views electronic dissemination of information as particularly beneficial to individual investors (e.g., SEC 2017), as these investors frequently use the internet, including social media, to obtain investment information (Shell 2015).

CEOs have economic incentives to build relationships with individual investors through social media disclosure channels. For example, recent regulation provides this investor group a voice in CEO compensation decisions (i.e., "say-on-pay" per the Dodd-Frank Act of 2010; U.S. House of Representatives 2010). Although say-on-pay votes are nonbinding, evidence shows that firms modify CEO compensation in response to investor recommendations (Balsam, Boone, Liu, and Yin 2016). CEOs that are viewed more favorably by investors are rewarded with positive say-on-pay votes and, ultimately, increased compensation (Kaplan, Samuels, and Cohen 2015).[1]

Although prior research shows that the disclosure channel has both direct and indirect effects on how investors process and perceive firm disclosures (Kelton, Pennington, and Tuttle 2010), little is known about the theoretical underpinnings explaining such results. Given the rapid speed at which technology changes, understanding the theoretical mechanisms underlying investors' use of different disclosure channels is critical. As noted by McFarland and Ployhart (2015, 1672–1673), "a theory-based understanding of social media offers a scientific foundation that is useful regardless of what platforms are popular at the moment." Accordingly, we develop a theoretical mediation model based on social capital theory that predicts that social media disclosure channels lead investors to perceive enhanced feelings of connectedness (i.e., social capital) with the CEO, resulting in more positive judgments of the CEO. We use a CEO's use of Twitter ("CEO tweets") as the context to examine our model, as research shows Twitter is a preferred social media for corporate communication (Zhou, Lei, J. Wang, Fan, and A. Wang 2015).

Social capital is the system of connections, relationships, and resources embedded in a social network (Nahapiet and Ghoshal 1998). Prior research supports that social capital is a three-dimensional construct (i.e., structural, cognitive, and relational capital) and that each dimension has important independent and interrelated effects (Sun, Fang, Lim, and Straub 2012). Structural capital is created by the physical connections and interactions between individuals in a social system and leads to the development of cognitive and relational capital. Cognitive capital represents the shared goals and common understanding developed from these interactions (i.e., structural capital), and relational capital represents the relational resources created through these interactions, such as social trust (Nahapiet and Ghoshal 1998).

We draw from prior social capital research and suggest that investors viewing a CEO's tweets will perceive enhanced social capital with the CEO, as compared to those reading the same information on a website. Individuals often perceive information exchanges via Twitter as being similar to face-to-face interactions, even though the interactions are text-based and frequently one-sided (Lee and Jang 2013). Communications research refers to this phenomenon as parasocial interaction, wherein a media user develops the illusion of intimate interaction with a media figure, even though they have never met in person (Horton and Wohl 1956; Rubin, Perse, and Powell 1985; Lee and Jang 2013). Based on this research, we use parasocial interaction as a measure of perceived structural capital and predict that investors who view a CEO's tweets will perceive higher levels of parasocial interaction than those viewing the same disclosures on a website. Parasocial interaction leads to feelings of liking and friendship with the media figure and positively influences judgments of the media figure and subsequent behaviors (Rubin and Step 2000; Labrecque 2014). Accordingly, we predict that parasocial interaction will be positively associated with both shared goals (i.e., cognitive capital) and social trust (i.e., relational capital) and will mediate the association between disclosure channel (Twitter versus website) and shared goals and social trust. Finally, as social capital is a valuable resource with persuasive effects (e.g., Belliveau, O'Reilly, and Wade 1996; He, Qiao, and Wei 2009; Hau, B. Kim, Lee, and Y. Kim 2013), we expect shared goals and social trust to be positively associated with investor recommendations for CEO compensation.

Although prior research in other disciplines finds a positive association between social media use, parasocial interaction, and individual judgments in the context of politicians (Lee and Oh 2012; Lee and Jang 2013) and actors (Lee and Jang 2013), it

[1] Although CEOs also have incentives to influence investors' judgments of the firm (performance expectations, investment likelihood, etc.), Siegel and Brockner (2005) find that CEOs' actions may have opposing effects on CEO compensation and firm value.

is unclear whether similar results will be found in the context of CEO-investor communications for at least two reasons. First, CEOs often make strategic decisions regarding corporate disclosures, and evidence suggests that investors are aware of these decisions. Such awareness may affect whether and how investors are influenced by variations in disclosure channel (Nelson and Rupar 2015). Second, investors may not view social media as an appropriate channel for CEO disclosure of financial information, which may lead investors to discount the information and/or question the credibility of the CEO and the disclosure. This "old media heuristic" suggests that individuals develop norms and expectations based on prior experiences with more traditional media (e.g., web-based, paper-based) and will negatively evaluate information encountered in a new or different media or channel (Sundar 2008).

To test our model, we conduct an experiment using individual investors.[2] Participants were randomly assigned to one of two conditions. Both conditions were provided identical information about a hypothetical firm and the firm's CEO followed by a series of press releases from the company in a web-based format. The initial press releases provided information about prior firm performance and the CEO's goals (Elliott, Hodge, and Sedor 2012). The final press release from the company announced employee layoffs, which prior research shows impacts decisions regarding CEO compensation (Henderson, Masli, Richardson, and Sanchez 2010). Participants then received a disclosure from the CEO discussing the layoffs, which included the disclosure channel manipulation. In order to isolate effects due to disclosure channel, one condition received the disclosure in the same web-based format as the prior announcements and the other condition received the identical information in a series of tweets. Finally, participants responded to a series of social capital questions and made a compensation recommendation for the CEO.

Overall, our results support our research model and provide evidence that social capital mediates the relationship between a CEO's use of Twitter and investors' recommendations for CEO compensation. Specifically, investors who view CEO tweets perceive higher levels of parasocial interaction, as compared to those reading identical information in a web-based format. Parasocial interaction is positively associated with both shared goals and social trust and mediates the associations between disclosure channel and shared goals and social trust. In addition, the relationship between parasocial interaction and compensation recommendations is mediated through social trust, while the effect of shared goals on compensation recommendations is indirect and mediated by social trust. Taken together, our results provide strong evidence that the use of Twitter (versus websites) for information disclosure leads to investor perceptions of enhanced social capital with the CEO, and in turn is positively associated with investor recommendations for CEO compensation.

Our study makes several contributions. First, by using an experiment, we are able to hold constant disclosure characteristics such as valence, content, and source, and identify the psychological processes underlying investor responses to corporate social media, which are not available in archival data. We thus answer calls from academics (Debreceny 2015; Moffitt, Richardson, Snow, Weisner, and Wood 2016) for experimental research that triangulates findings from archival studies investigating the market-level consequences of social media (Blankespoor et al. 2014; Du and Jiang 2015; Lee, Hutton, and Shu 2015). We also contribute to the larger body of research examining the influence of disclosure channels (Kelton et al. 2010) and disclosure characteristics (e.g., Janvrin, Pinsker, and Mascha 2013; Kelton and Murthy 2016) on investor judgments and decisions. Prior research shows that social connections between the CEO and board members affect investor decisions (e.g., Kaplan et al. 2015) and firm outcomes (e.g., Clune, Hermanson, Tompkins, and Ye 2014). To our knowledge, we are the first to measure investor perceptions of social connections (i.e., social capital) with the CEO, due to variations in disclosure channel, and the resultant influence on investor judgments. Our results thus contribute to the larger body of research on social capital.

Results should also inform regulators and firms of how investor judgments are influenced when CEOs use Twitter to make disclosures. Our evidence that CEOs can use social media to build social capital with investors, and ultimately impact compensation judgments, should be of particular concern to regulators.

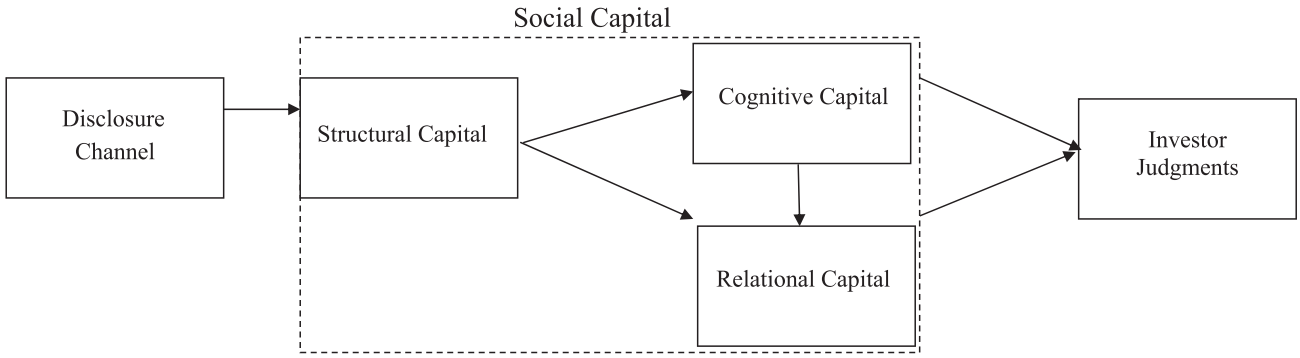
II. BACKGROUND AND HYPOTHESES DEVELOPMENT

Social Media and Corporate Disclosure

Social media are internet-based technologies that enable the creation, sharing, and exchange of information via online networks (Ellison, Steinfield, and Lampe 2007). Social media, such as Twitter, enable firms to communicate directly with a larger group of stakeholders in a timelier nature, as compared to alternative disclosure channels (e.g., paper, corporate website; Lee et al. 2015). In addition, social media use reduces investors' acquisition costs since disclosures are sent directly to investors instead of requiring investors to search for the information themselves (i.e., "push" versus "pull" technology; Blankespoor et al. 2014). Standard setters have approved the use of social media for information disclosure. For instance, the SEC announced

[2] Approval to conduct this experiment was granted by the author's institution.

FIGURE 1
Theoretical Framework



that firms may use social media to disseminate important corporate information in compliance with Regulation Fair Disclosure as long as investors are informed as to which sites will be used (SEC 2013). The SEC maintains that corporate social media use will improve the overall information environment (SEC 2013).

Given these proposed benefits, social media have become an integral component of firms' and CEOs' communication strategies. According to recent studies, approximately 49.5 percent of publicly traded U.S. firms use social media (Zhou et al. 2015), and approximately 40 percent of Fortune 500 CEOs maintain social media accounts (CEO.com 2016). These "social CEOs" (Weber Shandwick 2014) report using social media not only to disseminate important information to investors, but also to improve their own reputation and that of the firm, to give the firm a "human face," to build relationships with employees and the news media, and to improve business results. CEOs, however, also acknowledge potential risks associated with social media use, including unintentional errors in message content and regulatory concerns (Weber Shandwick 2014).

Despite the growth of social media use by CEOs, there is limited evidence regarding how such use affects individual investors. Findings from archival research suggest economic benefits associated with firms' use of social media (Du and Jiang 2015; Lee et al. 2015) and that social media disclosures are informative to investors (Blankespoor et al. 2014). For example, Blankespoor et al. (2014) examine information technology firms' use of Twitter to disseminate hyperlinks to press releases and find that such use is associated with reductions in information asymmetry. Du and Jiang (2015) find a positive association between the social media presence of a sample of S&P 1500 firms and firm performance. Saxton and Anker (2013) report reductions in information asymmetry associated with financial blog posts.

Although these studies demonstrate benefits associated with corporate social media use at the aggregate level, the influence of social media disclosures on individual investors is largely unexplored. In this study, we draw on social capital theory from management and suggest a relationship-centric explanation for social media effects. Specifically, we develop a theoretical mediation model that proposes social capital as the underlying mechanism explaining how a CEO's tweets influence investors' judgments. Figure 1 presents the theoretical framework.

Social Capital Theory

Social capital is "the sum of the actual and potential resources embedded within, available through, and derived from the network of relationships possessed by an individual or social unit" (Nahapiet and Ghoshal 1998, 243). Social capital develops from social interactions and the trust, coordination, and cooperation among individuals in a social network (Putnam 1995; Nahapiet and Ghoshal 1998). Nahapiet and Ghoshal (1998) classify social capital into three dimensions: structural capital, cognitive capital, and relational capital. Structural capital is "the overall pattern of connections between actors—that is, who you reach and how you reach them" (Nahapiet and Ghoshal 1998, 244). Structural capital is created by the physical ties and/or interactions between individuals in a social system. Cognitive capital is the shared goals among members in a social system (Nahapiet and Ghoshal 1998), who may otherwise have divergent goals. Cognitive capital is developed via structural capital and represents "the degree to which network members share a common understanding and approach to the achievement of network tasks and outcomes" (Inkpen and Tsang 2005, 153). Relational capital is the assets created and leveraged through network interactions, including social trust and expectations and obligations. Relational capital is impacted by both structural capital and cognitive capital and represents the relational resources embedded in the social network (Nahapiet and Ghoshal 1998).

Although social capital theory was originally examined in the context of face-to-face interactions, recent research shows that social capital also develops from online interactions (Wasko and Faraj 2005; Ellison et al. 2007; Hofer and Aubert 2013).

For example, Wasko and Faraj (2005) find that individuals participating in an electronic network of practice build social capital with other network participants. Social capital development also occurs among teams communicating with synchronous text messaging (Robert, Dennis, and Ahuja 2008), employees using an intraorganizational microblog (Sun and Shang 2014), and undergraduate student Facebook users (Ellison et al. 2007).

A fundamental tenet of social capital theory is that social interactions create valuable resources that influence individual attitudes and behaviors (Nahapiet and Ghoshal 1998). Research in a variety of contexts shows that social capital is used to influence others and achieve desired results. For example, Belliveau et al. (1996) show that a CEO's social capital, relative to the social capital of the compensation committee chairperson, is positively associated with CEO compensation. Information systems research shows that social capital positively affects individuals' intentions to share information via knowledge management systems (He et al. 2009; Hau et al. 2013).

In this study, we use social capital theory to explain how social media disclosure channels influence investor judgments. Prior research shows that each of the three social capital dimensions directly or indirectly facilitates the exchange of resources and/or desired behaviors and that the dimensions are interrelated (Nahapiet and Ghoshal 1998; Sun et al. 2012). Several studies document that structural capital leads to the development of both cognitive capital and relational capital. Further, these studies show that the impact of structural capital on outcomes is both direct and indirect through cognitive and relational capital (Tsai and Ghoshal 1998; Sun et al. 2012; Sun and Shang 2014). For example, Chow and Chan (2008) find that social networks and shared goals are both positively associated with employees' attitudes toward knowledge sharing. Several studies (Levin and Cross 2004; Hau and Kim 2011) find that measures of trust (relational capital) mediate the effect of strong ties (structural capital) on knowledge sharing. Sun et al. (2012) examine the role of social capital in user satisfaction with information technology service delivery. Results from Sun et al. (2012) show that cognitive and relational capital are positively associated with user satisfaction and also mediate the impact of structural capital on user satisfaction. Sun and Shang (2014) find that shared vision (cognitive capital) and trust (relational capital) mediate the effect of social ties (structural capital) on organizational social media use. Accordingly, we consider the three interrelated dimensions of social capital in the context of social media disclosures as shown in Figure 1.

CEO Tweets and Social Capital

We use Twitter as the social media disclosure channel to empirically examine our model. Research shows Twitter is a preferred social media platform for corporate communication (Zhou et al. 2015; CEO.com 2016) and thus provides a relevant and realistic setting for examining our theoretical model. We first investigate the effect of CEO tweets on structural capital. Structural capital is generated by the structure of the social network, including the location and frequency of interactions among individuals in the network, and will thus likely differ depending on the CEO's disclosure channel. Although prior research typically examines *actual* variations in structural capital, in this study we explore investor *perceptions* of structural capital in the context of CEO-investor communications.[3] Accordingly, we use parasocial interaction from communications and social media research as a measure of perceived structural capital.

Individuals often perceive interactions via Twitter to be similar to face-to-face interactions, even though tweets are frequently text-based and one-sided (Lee and Jang 2013). Communications and media researchers refer to this phenomenon as parasocial interaction. Described as an "illusionary experience" (Labrecque 2014, 135) and "intimacy at a distance" (Horton and Wohl 1956, 215), parasocial interaction is a media user's perception of personal contact and interpersonal interaction with media figures in the absence of any direct interactions (Horton and Wohl 1956; Rubin et al. 1985; Lee and Oh 2012). As a result of the media encounter, the user believes the media figure is communicating directly with him or her and feels as though they have actually met in person and therefore feels more connected to that media figure. Parasocial interaction has been explored across a variety of media channels primarily in the entertainment context. For example, individuals experience parasocial interaction with radio talk show hosts (Rubin and Step 2000), soap opera characters (Perse and Rubin 1989), actors on reality-based television shows (Nabi, Stitt, Halford, and Finnerty 2006), and video game avatars (Jin and Park 2009). Importantly, these studies show that parasocial interaction can exist in the absence of any *actual* interactions—i.e., media users only need to think or feel as though they have interacted with the media figure to experience the feelings of connectedness.

Recent research investigates parasocial interaction in the context of social media use providing some evidence that perceptions of interaction are intensified in the social media context. For example, Colliander and Dahlén (2011) find that

[3] We note that most social capital studies use survey methods to examine the consequences, as opposed to the antecedents, of social capital. Survey respondents in these prior studies report considerable variation with respect to their locations in social networks and the frequency of their interactions with others in the network (e.g., Chow and Chan 2008; Sun et al. 2012). As discussed in Section III, we hold constant the actual structure of the network—i.e., all participants assumed the role of a current investor in the company and all participants received the same number of communications from the CEO. We only vary disclosure channel and thus are able to isolate differences in perceptions of structural capital due to channel alone.

parasocial interaction is higher for individuals reading an article on a blog than those reading the same article in an online magazine. Lee and Jang (2013) show that individuals viewing a famous actor's Twitter page report higher levels of parasocial interaction than those viewing the same information on a news website. Lee and Shin (2014) examine voter responses to a politician's disclosures on Twitter and to the same information reported in an online newspaper article. Results show that individuals reading Twitter often experience higher parasocial interaction than those reading the web-based article.[4]

In summary, due to the social nature of the communication channel, individuals perceive information exchanges via social media as similar to face-to-face and interpersonal interactions where a stronger personal connection exists. We thus predict that investors that view CEO tweets will perceive higher levels of parasocial interaction (i.e., structural capital) than those that view the same information in a web-based format. Formally stated:

H1: Individual investors that view a CEO's disclosures on a social media disclosure channel [Twitter] experience higher levels of parasocial interaction than those that view the same disclosure on a corporate website.

Parasocial interaction is a key element in the model because it captures the essence of the social media disclosure channel experience—the individual feels a greater social connection than other channels. Parasocial interaction has a persuasive influence such that it leads to judgments of the communication and the communicator that are more positive (Rubin and Step 2000). Accordingly, we expect parasocial interaction to be positively associated with both cognitive capital and relational capital. Taken together, we expect parasocial interaction to mediate the relationship between CEO tweets and both cognitive capital and relational capital. Following prior research (Chow and Chan 2008; Hau and Kim 2011; Sun et al. 2012; Hau et al. 2013), we examine shared goals and social trust as the primary dimensions of cognitive capital and relational capital, respectively, as these measures are most relevant to the current context.

According to Inkpen and Tsang (2005, 153), "Shared goals represent the degree to which network members share a common understanding and approach to the achievement of network tasks and outcomes." Social interactions facilitate the development of a shared understanding and a common set of goals among members in a social network who may otherwise have divergent goals. For example, Rubin and Step (2000) show that parasocial interaction led talk radio listeners to develop attitudes and behaviors consistent with those of the host. Lee and Jang (2013) report that individuals reading a politician's social media page were more likely to vote for the candidate than those reading information from a news website. Although agency theory suggests that CEOs and investors possess different incentives and thus may not have similar goals or outcomes (Jensen and Meckling 1976), higher levels of parasocial interaction will lead investors to perceive that they share common goals with the CEO. We thus expect the increase in parasocial interaction from CEO tweets to enhance investor perceptions of shared goals with the CEO. Formally stated:

H2a: Parasocial interaction is positively associated with individual investors' perceptions of shared goals with the CEO and mediates the association between disclosure channel and shared goals.

Trust is an important antecedent to decisions involving risk, such as investment-related decisions (Elliott et al. 2012), and is particularly relevant in the current study. Trust is a vulnerable state in which an individual decides to rely on others. Trust develops based on one's expectations that others will not behave opportunistically but will act in a dependable and socially appropriate manner (Gefen, Karahanna, and Straub 2003). Social trust develops from social interactions and communication (Chow and Chan 2008). Interactions among members of a social network provide opportunities for information sharing and relationship development, which reduce uncertainty regarding others' intentions and behaviors. In addition, social interactions lead to the development of a positive attitude and sense of belonging to others in the social network, resulting in social trust (Hau and Kim 2011). Thus, we expect the increase in parasocial interaction from CEO tweets to enhance investor perceptions of social trust in the CEO. Formally stated:

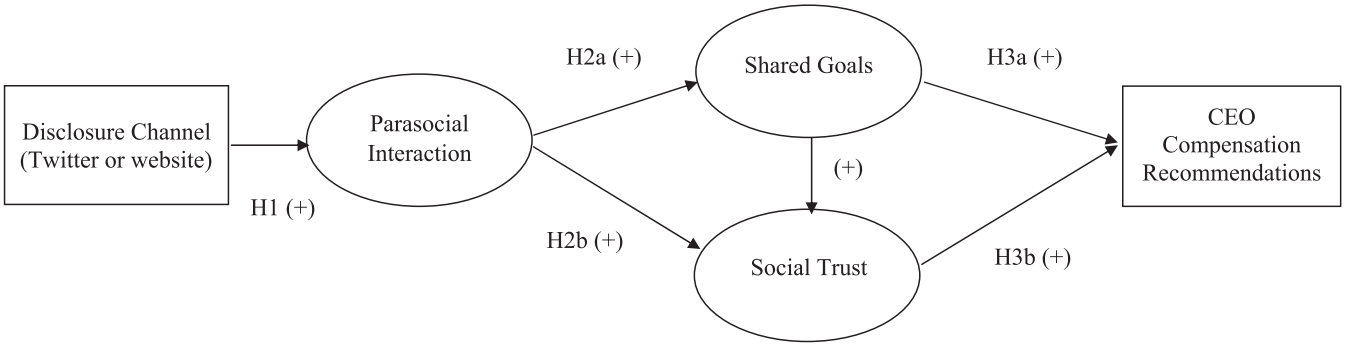
H2b: Parasocial interaction is positively associated with individual investors' perceptions of social trust in the CEO and mediates the association between disclosure channel and social trust.

The Effect of Social Capital on Investor CEO Compensation Recommendations

Finally, we explore the impact of social capital on investors' judgments. Specifically, we expect that investors' perceptions of shared goals and social trust with the CEO will positively influence their recommendations for CEO compensation. Issues involving CEO compensation have recently received increased attention from regulators due to perceptions that CEO pay is

[4] Importantly, these studies vary disclosure channel (e.g., social media versus websites) in addition to other characteristics, such as message source (e.g., original source versus third party; Lee and Jang 2013; Lee and Shin 2014) and message personalization (Lee and Oh 2012), which likely will impact parasocial interaction. It is thus unclear from prior research whether variations in parasocial interaction would occur due to channel differences alone.

FIGURE 2
Research Model



unfair and excessive. Specifically, the U.S. Congress passed the Dodd-Frank Wall Street Reform and Consumer Protection Act (U.S. House of Representatives 2010), which mandates the following: (1) companies must provide investors a say-on-pay, which is a nonbinding vote on executive compensation packages, and (2) companies must disclose the ratio of CEO compensation to the median compensation of employees in order to better inform investors in their say-on-pay votes.

Accounting research on say-on-pay votes concerning individual investors is sparse (Obermann and Velte 2018). In general, research finds that these votes are important to firms and CEOs. For example, even though say-on-pay votes are nonbinding, firms tend to make decisions consistent with such votes (Balsam et al. 2016; Ferri and Oesch 2016; Kimbro and Xu 2016). In addition, investor perceptions of the CEO (e.g., reputation; Kaplan et al. 2015) and the firm (Kaplan and Zamora 2018) influence say-on-pay votes. Ferri and Oesch (2016) provide evidence that management credibility determines the extent to which management yields influence over investor votes on say-on-pay issues. Given these findings, investor judgments regarding CEO compensation provide an interesting context in which to examine social media effects on social capital development.

As previously discussed, social capital influences the attitudes and behaviors of those in a social network (Belliveau et al. 1996; He et al. 2009; Hau et al. 2013). In the context of social media, van Noort, Antheunis, and van Reijmersdal (2012) find that social relationships between the sender and recipient of a viral social media marketing campaign positively influence the recipient's attitudes toward the brand and the marketing campaign. In the accounting domain, Lee et al. (2015) examine a sample of consumer product recalls and find that a firm's use of social media attenuates the negative price reaction to the recall announcement. Lee et al. (2015) attribute the results to improvements in credibility and reputation gained through firms' social media communications, although they are unable to support this reasoning with archival data.

Accordingly, we expect that higher levels of shared goals and social trust, due to enhanced parasocial interaction from CEO tweets, will lead to recommendations for CEO compensation that are more positive. We thus predict that both shared goals and social trust are positively associated with CEO compensation recommendations. Formally stated:

H3a: Individual investors' perceptions of shared goals with the CEO are positively associated with their CEO compensation recommendations.

H3b: Individual investors' perceptions of social trust in the CEO are positively associated with their CEO compensation recommendations.

Figure 2 presents the research model. Prior social capital research reports an established positive and direct effect from cognitive capital to relational capital (Tsai and Ghoshal 1998; Sun et al. 2012). We include the path in our model to control for this effect but do not hypothesize the relationship in accordance with prior research.

III. METHOD

Participants and Procedures

We utilized a single factor between-participants design to test our hypotheses. We manipulated disclosure channel and, specifically, whether participants viewed a letter from the CEO on the company's corporate website or on Twitter. All information content was held constant across both conditions. We used Amazon Mechanical Turk (AMT) to obtain 179

TABLE 1

Participant Demographic Statistics

	Mean[a] (Std. Dev.)	Count[a] (Percentage)
Investing Experience	4.72 (5.78)	
Work Experience	14.17 (8.83)	
Accounting Courses	2.66 (17.03)	
Finance Courses	1.83 (8.06)	
Age		
18–25 years old		29 (16.5%)
26–35 years old		78 (44.3%)
36–45 years old		46 (26.1%)
46–55 years old		21 (11.9%)
56–65 years old		2 (1.2%)
Twitter Use Intensity	17.29 (11.92)	

[a] Investing experience is measured by participants' self-reported years of investing experience at a personal level. Work experience is measured by the number of years of professional work experience. Participants self-reported the number of upper-level accounting and finance courses they have taken. Twitter use intensity is measured by the sum of participants' responses to the following four items using an 11-point scale anchored by (1) Strongly Disagree and (11) Strongly Agree: Twitter is part of my everyday activity; I would be sorry if Twitter shut down; I use Twitter to get useful information; and I am proud to tell people I am on Twitter (Ellison et al. 2007).

investor participants for our study in exchange for a $1.50 payment.[5] Participants were prescreened before completing the task and only those individuals who considered themselves native English speakers and had previously owned common stocks or mutual funds or planned to purchase them in the future were permitted to complete the task.

Table 1 presents the participant demographic statistics. On average, participants reported 4.72 years of personal investing experience, 14.17 years of work experience, and having taken 2.66 (1.83) accounting (finance) courses. Approximately 44 percent (26 percent) of our participants were between 26–35 (36–45) years old. Thus, these participants have the requisite knowledge to complete our experimental task (Libby, Bloomfield, and Nelson 2002).

There was no significant difference in participant demographics across disclosure channel conditions. To control for any effects due to participants' use of and familiarity with Twitter on the hypothesized relationship between disclosure channel and parasocial interaction, we conducted a between-subjects ANCOVA with Twitter use intensity as a covariate.[6] Results show a significant difference in parasocial interaction between groups ($p = 0.032$ one-tailed). Therefore, we conclude that differences in parasocial interaction between groups are robust to any effects due to participants' Twitter usage.[7]

Participants received an online case scenario involving a hypothetical company, a provider of personal computing peripheral devices. Participants were instructed to assume the role of a current investor in the company and were given background information about the company and the CEO. Participants then reviewed eight press releases about the company in

[5] See Farrell, Grenier, and Leiby (2017) and Rennekamp (2012) for discussions of the use of AMT for experimental accounting research. There were no duplicate IP addresses in the data indicating that no participant attempted to complete the study more than one time.

[6] Using an 11-point scale anchored by (1) Strongly Disagree and (11) Strongly Agree, participants indicated their agreement with the following four items: Twitter is part of my everyday activity; I would be sorry if Twitter shut down; I use Twitter to get useful information; and I am proud to tell people I am on Twitter (Ellison et al. 2007).We measured participants' Twitter use intensity with the sum of the four items.

[7] We also included Twitter use intensity as a covariate in the SEM analysis. Twitter use intensity was not significant and the results were quantitatively similar to the results in Figure 3 and Table 3. Therefore, we do not include Twitter use intensity in any of the analyses.

a web-based format. The first seven press releases conveyed positive news (e.g., new product lines, meeting analyst earnings expectations) in order to provide a baseline of information regarding the prior goals and performance of the company and the CEO. The final press release announced the layoff of a number of employees over the following year. After viewing the press releases, we manipulated whether the participants received a disclosure from the company's CEO discussing the layoffs (1) on the company's corporate website, or (2) on Twitter (Appendix A).[8] We held constant the information content across disclosure channel conditions and the timing of the information by presenting all tweets on the same page and at the same time. After viewing all the information, participants responded to a series of questions.

Measures

Appendix A presents the measurement items. Parasocial interaction was measured with six items adapted from Lee (2013). Three items were subsequently removed from the analysis as described in the next section. We measured shared goals and social trust using items adapted from prior research (Chiu, Hsu, and Wang 2006; Chow and Chan 2008; Hau and Kim 2011). Participants were informed that the CEO's 2012 salary was $1,750,000, which was in line with industry averages for similarly sized companies, and that the compensation committee of the board of directors was considering a 2013 base salary increase to $2,000,000. To measure CEO compensation recommendation, participants indicated what they think the CEO's base salary should be for 2013 using a 13-point scale anchored by (1) less than $1,500,000 and (13) more than $2,000,000, with each point increasing in $50,000 increments (Siegel and Brockner 2005).[9]

We carefully designed the CEO compensation information and held the level of compensation constant across experimental conditions so as not to introduce unnecessary noise into the experiment. The CEO compensation amounts used in the experimental task were consistent with those used in prior research (Kaplan et al. 2015; Kaplan and Zamora 2018). In addition, accounting researchers with expertise in CEO compensation reviewed the instrument for reasonableness. We measured investor perceptions of the fairness of CEO compensation with the sum of the following four items using an 11-point scale anchored by (1) strongly disagree and (11) strongly agree (Kaplan et al. 2015): (1) Total compensation for Dan Taylor, Advanced's CEO, is fair to shareholders; (2) Total compensation for Dan Taylor, Advanced's CEO, is excessive (RC); (3) The process used by the compensation committee to compensate Dan Taylor is fair to shareholders; and (4) The compensation committee is biased in favor of the CEO, Dan Taylor (RC). Perceptions of compensation fairness did not differ significantly across experimental conditions (Twitter = 19.82 versus website = 18.67; p-value = 0.415).

IV. RESULTS

We tested the hypotheses using structural equation modeling (SEM), SPSS AMOS Version 22. We selected SEM as the most appropriate method due to its ability to simultaneously estimate a mediated model that includes latent variables. The model includes three latent variables (parasocial interaction, shared goals, and social trust), one manifest variable (CEO compensation recommendation), and one manipulated variable (disclosure channel). Disclosure channel is coded as 1 for the Twitter condition and as 0 for the website condition.

Measurement Model

The measurement model consists of the three latent variables previously described. The parasocial interaction variable has three indicators,[10] shared goals consists of three indicators, and social trust has four indicators. Each variable has one indicator that was set at unity. Table 2 presents the analysis of the reliability and convergent validity of the measurement model. Cronbach's alpha for each variable is above 0.90 suggesting good internal reliability. Confirmatory factor analysis was conducted to test the measurement model and evaluate convergent and discriminant validity. Goodness of fit indices indicate

[8] To ensure the external validity of the disclosure channel manipulations, the Twitter condition used a series of short tweets, while the website condition presented the same information in paragraph form. An alternative explanation for the observed disclosure channel effects is that the Twitter condition was more readable than the website condition (e.g., Rennekamp 2012; De Franco, Hope, Vyas, and Zhou 2015). We conducted a follow-up study to rule out this possible alternative explanation using 46 executive M.B.A. students from a private university in the southeastern U.S. Following the procedures previously described, participants were provided the same initial press releases and then viewed the CEO letter either on the corporate website or on Twitter. Participants then assessed the readability of the disclosure by indicating how easy or difficult it was to read (understand) the information from the CEO using an 11-point scale with (1) very difficult, (6) neutral, and (11) very easy (Tan, Wang, and Zhou 2014). There were no significant differences in readability (means = 7.74 [Twitter] versus 8.74 [website], t = 1.314; p = 0.196) or understandability (means = 7.17 [Twitter] versus 8.26 [website], t = 1.289; p = 0.204) across the disclosure channel conditions thus ruling out this possible alternative explanation.

[9] We counterbalanced the order that participants responded to the social capital measures and the compensation judgment. Order is not significant when included in the hypotheses tests, thus it is excluded from further discussion.

[10] Lee and Oh (2012) suggest six indicators; however, in our sample, three of those items produced high residual covariances and were therefore removed from the model (Hooper, Coughlan, and Mullen 2008).

TABLE 2

Scale Reliability and Convergent Validity

Construct[a]	Mean (Std. Dev.)	Cronbach's alpha	Composite Reliability	AVE	Item	Factor Loading
Parasocial Interaction	15.62 (7.81)	0.909	0.910	0.772	PSI 1	0.869
					PSI 2	0.929
					PSI 3	0.836
Shared Goals	20.08 (7.75)	0.943	0.943	0.847	SG 1	0.902
					SG 2	0.920
					SG 3	0.938
Social Trust	24.34 (10.31)	0.940	0.942	0.803	ST 1	0.886
					ST 2	0.925
					ST 3	0.816
					ST 4	0.951

[a] Parasocial interaction is measured using the sum of three items, shared goals is measured using the sum of three items, and social trust is measured using the sum of four items (see Appendix A). Each item is measured on an 11-point scale ranging from strongly disagree to strongly agree.

that the measurement model provides a good fit to the data. The SRMR, RMSEA, NFI, TLI, and CFI are 0.018, 0.050, 0.976, 0.990, and 0.993, respectively, which are all well within the recommended cut-off values (Hu and Bentler 1999). In addition, the Chi-squared/df ratio is 1.44 (< 3 is good fit). Convergent validity was evaluated using the factor loadings (> 0.60), the average variance extracted (AVE) (> 0.50), and the composite reliability (> 0.70) (Hair, Black, Babin, and Anderson 2010). All items exceed the criteria for convergent validity (see Table 2). Discriminant validity was assessed by the Fornell-Larcker test (Fornell and Larcker 1981), which establishes that the square root of the AVE for each construct should exceed any inter-construct correlations. The inter-construct correlation between parasocial interaction and social trust is 0.85, parasocial interaction and shared goals is 0.72, and shared goals and social trust is 0.86. The square root of the AVE for each construct (parasocial interaction = 0.88, shared goals = 0.92 and social trust = 0.90) exceeds these correlations and thus establishes discriminant validity. Overall, the measurement model provides a good fit to the data.

Structural Equation Model

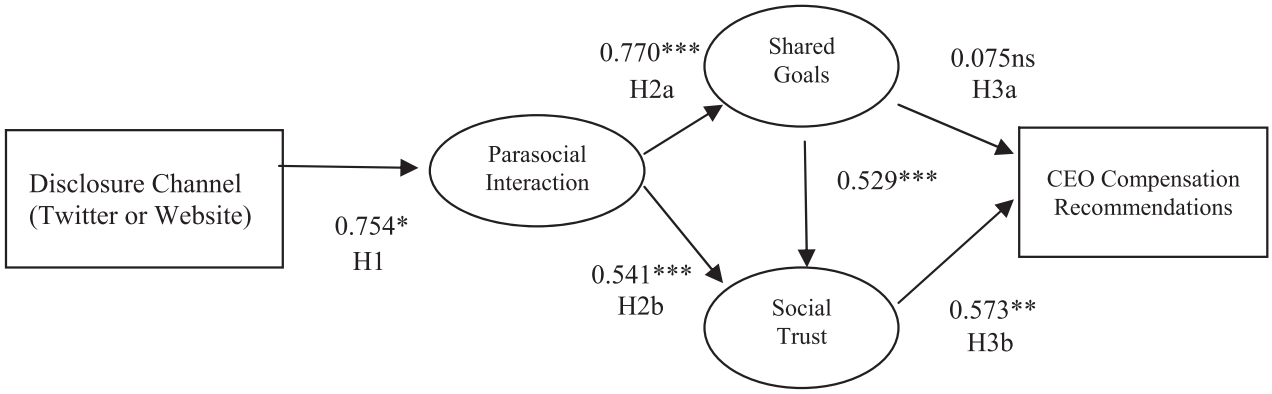
Figure 3 presents the structural equation model and includes the path coefficients and fit indices.[11] Overall, the results for the structural model show a good fit to the data. The SRMR, RMSEA, NFI, TLI, and CFI are 0.024, 0.046, 0.966, 0.987 and 0.990, respectively. The Chi-squared is 69.22 with 50 degrees of freedom and a ratio of 1.384. Table 3 presents regression weights, standard errors, critical ratios, and p-values.

H1 predicts that investors that view a CEO's disclosure on Twitter experience higher levels of parasocial interaction than those that view the same disclosure on a corporate website. As shown in Figure 3, the path from disclosure channel (Twitter or website) to parasocial interaction is positive and significant ($p < 0.05$), providing support for H1. Investors in the Twitter condition perceived higher parasocial interaction (mean = 16.88) than those in the web-based condition (mean = 14.38; $p = 0.017$).

H2a predicts that parasocial interaction is positively associated with investors' perceptions of shared goals with the CEO and mediates the association between disclosure channel and shared goals. H2b predicts that parasocial interaction is positively associated with investors' perceptions of social trust in the CEO and mediates the association between disclosure channel and social trust. As shown in Figure 3, the paths from parasocial interaction to both shared goals and social trust are both positive and significant (both $p < 0.001$), providing support for H2a and H2b. Taken together with results for H1, these findings suggest a mediating role of parasocial interaction. Specifically, we find that a CEO's use of Twitter is associated with perceptions of higher levels of parasocial interaction, leading to enhanced perceptions of shared goals and social trust. In other words, investors viewing a CEO's tweets experience feelings of greater social connectedness (i.e., social capital) with the CEO, as compared to those viewing the information on a corporate website.

[11] Our theory does not predict that disclosure channel will directly impact investor CEO compensation recommendations and results do not support the existence of a direct effect ($p = 0.441$). Rather, the theory supports a fully mediated model. As the purpose of this study is to understand the processes underlying investors' responses to CEO tweets, we focus our analyses on the mediating variables, as suggested by Libby et al. (2002).

FIGURE 3
SEM Analysis
Regression Weights (Unstandardized) and Fit Indices



*, **, *** Indicate one-tailed p-values of < 0.05, < 0.01, and < 0.001, respectively.
Chi-squared/df = 1.384
SRMR = 0.024
RMSEA = 0.046
NFI = 0.966
TLI = 0.987
CFI = 0.990

Finally, H3a and H3b predict positive associations between shared goals and social trust and investors' CEO compensation recommendations. As shown in Figure 3, the path from social trust to CEO compensation recommendation is positive and significant ($p = 0.004$), providing support for H3b. Perceived social trust is positively associated with CEO compensation recommendations. The path from shared goals to CEO compensation recommendation is not significant; therefore, H3a is not supported. In sum, these results support our research model and the mediating role of social capital in the influence of disclosure channel on investor judgments of CEO compensation.

TABLE 3
Regression Weights of the Research Model

	Estimate	Std. Error	Critical Ratio	p-values (one-tailed)
PSI ← Disclosure Channel	0.754	0.353	2.137	0.017
Shared Goals ← PSI	0.770	0.075	10.247	***
Social Trust ← PSI	0.541	0.072	7.490	***
Social Trust ← Shared Goals	0.529	0.066	8.050	***
Compensation ← Social Trust	0.573	0.215	2.667	0.004
Compensation ← Shared Goals	0.075	0.224	0.333	0.370
PSI 1 ← PSI	1.000			
PSI 2 ← PSI	1.157	0.067	17.390	***
PSI 3 ← PSI	1.078	0.074	14.579	***
Shared Goals 1 ← Shared Goals	1.000			
Shared Goals 2 ← Shared Goals	1.034	0.052	19.781	***
Shared Goals 3 ← Shared Goals	1.027	0.049	20.799	***
Social Trust 1 ← Social Trust	1.000			
Social Trust 2 ← Social Trust	1.030	0.053	19.425	***
Social Trust 3 ← Social Trust	0.871	0.059	14.764	***
Social Trust 4 ← Social Trust	1.005	0.048	20.782	***

***Indicates a p-value < 0.001, one-tailed.

TABLE 4

PROCESS Analysis of Multiple Mediation—Indirect Effects of Disclosure Channel on CEO Compensation Recommendations

	Effect	Boot SE	Boot LLCI	Boot ULCI
Total	0.4805	0.2650	−0.0068	1.0392
DC → PSI → CR	0.1141	0.1511	−0.1077	0.5172
DC → PSI → SG → CR	0.0756	0.1087	−0.0710	0.3808
DC → PSI → ST → CR	**0.1442**	**0.1111**	**0.0050**	**0.5065**
DC → PSI → SG → ST → CR	**0.1168**	**0.0809**	**0.0088**	**0.3594**
DC → SG → CR	0.0246	0.0780	−0.0510	0.2977
DC → SG → ST → CR	0.0380	0.0681	−0.0674	0.2310
DC → ST → CR	−0.0327	0.0842	−0.2480	0.1095

Bootstrap confidence intervals that do not contain 0 are evidence of significant indirect effects (Hayes 2012) and noted in bold font.
DC = Disclosure Channel.
PSI = Parasocial Interaction.
SG = Shared Goals.
ST = Social Trust.
CR = CEO Compensation Recommendation.

Robustness tests of our primary SEM analysis were performed to provide additional support for the mediation paths of our model. We conduct mediation tests using both the Baron and Kenny (1986) method and the PROCESS (Hayes 2012) method. Our research model contains three linked mediators (parasocial interaction, shared goals, and social trust). The traditional Baron and Kenny (1986) method does not consider multiple serially linked mediators simultaneously. Notwithstanding these limitations, we conduct the analysis and find results consistent with support for full mediation.[12] We also conducted the mediation analysis with PROCESS because it allows for the analysis of multiple mediators in one model as well as mediators that are linked serially. Similar to the Baron and Kenny (1986) method, the direct paths between independent variables and dependent variables are estimated as well as the mediator paths; however, the PROCESS analysis allows all paths to be estimated simultaneously in one model. The simultaneous estimation produces both direct and indirect effects similar to our main SEM analysis. The PROCESS analysis produces a bootstrap confidence interval for the indirect effects, which are interpreted as the mediating effects (Hayes 2012).

Results from PROCESS are presented in Table 4 and indicate significant indirect effects of two paths in our model: (1) the path from disclosure channel to parasocial interaction to social trust to CEO compensation recommendations (bootstrap confidence interval 0.0050 to 0.5065), and (2) the path from disclosure channel to parasocial interaction to shared goals to social trust to CEO compensation recommendations (bootstrap confidence interval 0.0088 to 0.3594).[13] These results add further support for the mediation predicted in our research model. In sum, our results show that when a CEO uses Twitter to disclose information instead of a corporate website, investors view the communication in a social context (i.e., enhanced social capital), leading to recommendations for CEO compensation that are more positive.

V. CONCLUSION

The SEC (2013) is encouraging corporate social media use to reduce information asymmetry and improve the overall information environment. CEOs, however, report using social media for more relational reasons, such as improving their own visibility and reputation (Weber Shandwick 2014). In this study, we draw upon communications and management theories to identify the psychological processes underlying investors' use of Twitter to acquire disclosures from a CEO. We develop a

[12] The Baron and Kenny (1986) method tests each mediator separately to determine if the mediator fully or partially mediates the path from the independent variable to the dependent variable. Mediation is established by first demonstrating significant relationships between the independent variable and the mediator and between the independent variable and dependent variable. Then, the relationship between the independent variable and the dependent variable is tested including the mediator. If the independent variable is no longer significant, full mediation is concluded. We use this test for each mediating variable in our research model (see Figure 2). Untabulated results support that parasocial interaction fully mediates the paths from disclosure channel to shared goals and social trust and that social trust fully mediates the path from parasocial interaction to CEO compensation recommendation. These results are consistent with the good fit of the SEM model.

[13] PROCESS produces bias-corrected confidence intervals for indirect effects. We use the recommended confidence interval of 95 percent. Confidence intervals excluding 0 are considered significant (Hayes 2012).

theoretical mediation model based on social capital theory that predicts that social media disclosure channels will lead investors to perceive enhanced feelings of connectedness (i.e., social capital) with the CEO, resulting in judgments of the CEO that are more positive. Specifically, our model predicts that Twitter, as compared to web-based, disclosures lead to enhanced perceptions of social capital, which in turn positively influence CEO compensation recommendations.

Results from an experiment show that investors viewing CEO disclosures via Twitter perceive higher levels of parasocial interaction as compared to those reading the same information on the firm's website. Parasocial interaction mediates the association between disclosure channel and investor perceptions of shared goals with and social trust in the CEO. That is, the increase in parasocial interaction from CEO tweets positively influences the development of shared goals and social trust. Higher levels of perceived social trust are positively associated with CEO compensation recommendations. The positive influence of shared goals on CEO compensation recommendation is indirect through social trust. Overall, we provide robust evidence that investors' use of Twitter to acquire CEO disclosures is positively associated with social capital, leading to investor recommendations for higher compensation for the CEO.

Our findings should be of critical importance to regulators, firms, and the investing community. Given the SEC's recent ruling on social media as a platform for information disclosure, results inform regulators of unanticipated consequences of a CEO's use of social media for information disclosure. These findings should be of particular interest given recent regulation providing investors a voice in CEO compensation decisions, providing additional incentives for CEOs to build relationships with investors. For example, dissenting say-on-pay votes are associated with changes to CEO compensation (Balsam et al. 2016; Kimbro and Xu 2016) and negative firm performance (Kimbro and Xu 2016). Our findings suggest that CEOs may use social media to influence investor perceptions of social capital with the CEO, leading to recommendations for higher CEO compensation. Our results also inform the investing community with evidence of the effects of using social media to acquire corporate disclosures, which should aid investors in determining what disclosures channels to use.

Our research contributes to several streams of literature. First, our results build upon the developing research stream on the use of social media for information dissemination by identifying social capital as the theoretical explanation for investor responses to using social media to acquire information. We show that when a CEO uses Twitter to disclose information instead of a corporate website, investors view the communication in a social context (i.e., enhanced social capital) and provide higher recommendations for CEO compensation.

Second, we extend findings in communications research of a positive association between social media use and parasocial interaction to the CEO-investor communication context. We find that the use of Twitter to disclose news is associated with higher levels of parasocial interaction, even when message source, content, and frequency are held constant. We thus provide a useful theoretical framework for future studies on new corporate disclosure channels and the resultant influence on investor judgments.

Third, we add to the social capital literature through the use of parasocial interaction as a measure of structural capital. Most prior social capital research examines *actual* variations in structural capital. We find that the effects of *perceptions* of structural capital (i.e., parasocial interaction) on cognitive and relational capital and, ultimately, individual judgments are similar to those of *actual* structural capital documented in prior research. We call for future research that explores other accounting settings (tax, auditing, managerial) in which parasocial interaction may influence outcomes.

Our study is subject to certain limitations, which are also areas for future research. We used Twitter instead of other social media as Twitter is a preferred social media platform for corporate communication (Zhou et al. 2015). Different social media utilize different functionality, and future research should examine how these characteristics influence investor decision making. In addition, one of the benefits of social media is the ability to facilitate interactive, two-way communications between firms and stakeholders (Lee et al. 2015). For example, tweets can be retweeted and individuals can comment on certain tweets. Although there may be both negative and positive consequences to interactivity in social media (Trinkle, Crossler, and Belanger 2015), we did not allow for interactivity in our experiment in order to maintain experimental control. The experimental materials also did not address the CEO's level of social media presence and influence (e.g., number of followers, number of likes, and retweets), which have been shown to influence credibility perceptions (Sundar 2008). Although we expect parasocial interaction to be even greater in the presence of interactivity (Lee and Shin 2012) and enhanced social media influence, these remain matters for future research.

Further, we designed the experimental task based on actual CEO disclosures (Twitter and website) and disclosed levels of compensation to maintain high levels of external validity within the experimental setting. While this choice resulted in the potential for a readability difference across conditions due to the inherent nature of the disclosure channels, this tradeoff with internal validity was important to enhance the generalizability of the results. Although additional analysis indicates these differences did not impact our results, we encourage research to further explore this matter. Future research should also explore whether and how other changes to compensation (e.g., higher salary increases, fixed versus variable pay) are influential on investor judgments of CEO compensation.

Additionally, our experimental design included a corporate layoff event—the content of which was consistent across conditions. Although we expect that our results would hold for other types of events and other types of disclosures (e.g., good versus bad news, quantitative versus qualitative), we encourage future research to investigate this matter. Finally, investors must choose to follow a CEO on Twitter in order to receive the CEO's tweets, and investors in this study were not provided such a choice. We leave examination of the extent to which such a choice effects results of this study to future research.

Despite these limitations, we provide robust evidence that the use of Twitter for information disclosure, as compared to a website, leads to investor perceptions of enhanced social capital with a CEO. The enhanced social capital in turn leads to more positive recommendations for CEO compensation.

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APPENDIX A

Excerpts from Experimental Materials

Twitter Disclosure Channel Condition



Website Disclosure Channel Condition



Measurement Items

The following items were measured using an 11-point scale anchored by (1) Strongly Disagree and (11) Strongly Agree:

Parasocial Interaction (Lee 2013)[a]

PSI 1. Dan Taylor makes me feel comfortable, as if I am with friends.
PSI 2. I see Dan Taylor as a natural, down-to-earth person.
PSI 3. I would like to meet Dan Taylor in person.

Shared Goals (Chow and Chan 2008)

SG 1. I believe that Dan Taylor and I agree on what is important regarding Advanced.
SG 2. I believe that Dan Taylor and I share the same ambitions and vision regarding Advanced.
SG 3. I believe that Dan Taylor and I are always enthusiastic about pursuing the collective goals and missions of the entire company.

Social Trust (Chiu et al. 2006; Hau and Kim 2011)

ST 1. I believe Dan Taylor will not take advantage of others, even when the opportunity arises.
ST 2. I believe Dan Taylor will always keep the promises he makes to others.
ST 3. I believe Dan Taylor behaves in a consistent manner.
ST 4. I believe Dan Taylor is truthful in dealing with others.

[a] The following three items were also collected and subsequently removed from the analysis as described in footnote 10: I look forward to hearing more news about Dan Taylor; Dan Taylor seems to understand the kinds of things I want to know; and I feel as if I have known Dan Taylor for a long time.

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